RESEARCH COMPILATION 2026

The LLM Stack

2026

Transformers, pretraining, RLHF, RLVR reasoning, test-time compute and interpretability: the complete 2017-2026 LLM pipeline in 38 papers.

Intelligence Research Atlas - Volume 03 of 6 -

Compiled 2026-10-25

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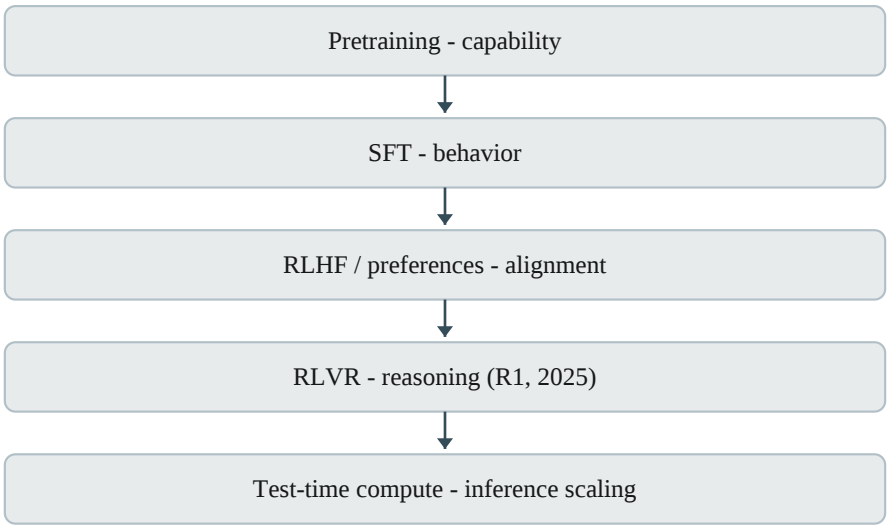
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LLM Architecture and Training

38 papers from Attention Is All You Need to the evaluation crisis, distilled into 12 findings covering what to build, how to align, and where reasoning breaks.

The LLM Stack - the through-line



Part I - Distilled Findings

F1 - The transformer won by parallelism, then by everything else

Implication for building intelligence: Architecture choices are compute-ecology decisions: the winning architecture is the one that best converts available hardware into capability. Any new architecture (domain 05) must answer: what does it parallelize?

F2 - Pretraining = capability; post-training = behavior

Implication for building intelligence: Budget accordingly: capability per dollar lives in pretraining data quality/quantity; behavior per dollar lives in post-training. Trying to teach facts via SFT is fighting the current.

F3 - RLHF is a three-actor system with known failure modes

Implication for building intelligence: Whatever preference machinery you build, monitor reward-model drift, cap KL to the reference policy, and prefer direct/preference losses (DPO/GRPO-family) unless you need online exploration.

F4 - Reasoning is trainable with verifiable rewards (RLVR) - the 2025 revolution

Implication for building intelligence: The default path to strong reasoning systems in 2026: base model + RLVR + verifiable environments. The bottleneck moved from algorithms to ENVIRONMENT DESIGN (problem generators, verifiers, reward shaping against hacking).

F5 - Test-time compute is a second scaling axis

Implication for building intelligence: Product/system design now has a tunable capability dial (latency/cost <-> quality). Compute-optimal INFERENCE planning (when to think long, when to answer directly) is a live engineering discipline.

F6 - Reasoning models have real limits: collapse past a complexity frontier

Implication for building intelligence: Do not assume reasoning models can handle arbitrary depth; decompose tasks, force tool use for exact steps (calculators, code), and evaluate on contamination-free procedural benchmarks (ARC-AGI class) before trusting scale narratives.

F7 - In-context learning is a real mechanism (induction heads), not a trick

Implication for building intelligence: ICL is the model's built-in learning algorithm - treat prompts as training data (order, format, and diversity matter statistically). Mechanistic knowledge lets you predict when ICL will fail (unfamiliar formats, conflicting patterns).

F8 - Models store more concepts than they have neurons (superposition)

Implication for building intelligence: Interpretability is now an engineering tool (steering, auditing, safety). Also the mechanistic root of scaling laws (domain 01) - capacity grows with interference-managed feature packing. If you build custom models, budget for feature-based monitoring.

F9 - Emergence claims need smooth metrics and mechanism checks

Implication for building intelligence: For capability forecasting (and safety planning): demand log-likelihood curves + mechanism, not exact-match cliffs. For research: if you find a cliff, look for the circuit underneath it.

F10 - Efficiency engineering is a first-class research area, not an afterthought

Implication for building intelligence: Open ecosystems win through efficiency (LLaMA recipe, DeepSeek economics). A capable small team can compete by engineering multiplier stacking: sparse architecture × low precision × kernel fusion × test-time compute allocation.

F11 - Knowledge lives badly in weights - retrieval separates knowing from storing

Implication for building intelligence: Architect systems as: weights = reasoning engine, external stores = facts, tools = exactness. This is also how agent memory systems (domain 05) are built in practice.

F12 - The evaluation crisis is the meta-problem of 2025–2026

Implication for building intelligence: When selecting or training models, weight contamination-resistant, procedurally-generated evaluations over leaderboards; build private evals reflecting your deployment distribution. Trust curves (smooth metrics over scale), not single numbers.

Part II - The Canon

38 landmark papers in chronological order. Full entries with links and keywords live in the corpus folder 03_llm-architecture-training/papers.md.

Year	Paper / Authors	Core idea	Why it matters
2017	**Attention Is All You Need** Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan Gomez, Łukasz Kaiser, Illia Polosukhin	Drop recurrence entirely; stacked multi-head self-attention + FFN with positional encodings, trained fully in parallel.	The substrate of every LLM; parallelism over sequence position is what made internet-scale training tractable.
2017	**Deep Reinforcement Learning from Human Preferences** Paul Christiano et al.	Learn reward from human preference comparisons instead of hand-crafted rewards.	The conceptual seed of RLHF, seven years before ChatGPT; also the seed of reward-hacking concerns.
2018	**BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding** Jacob Devlin et al. (Google)	Masked-language-model pretraining, bidirectional, then fine-tune per task.	Pretrain-then-adapt became the NLP paradigm; held every GLUE record on release.
2018	**Improving Language Understanding by Generative Pre-Training (GPT-1)** Alec Radford et al. (OpenAI)	Unidirectional (causal) LM pretraining + light task heads.	Chose generation over masking - the path that scaled into GPT-2/3 and everything since.
2019	**Language Models are Unsupervised Multitask Learners (GPT-2)** Alec Radford et al. (OpenAI)	1.5B parameters, web text (WebText) -> coherent multi-paragraph text, zero-shot task behavior.	First public evidence that scale alone induces task abilities without supervision - staged release foreshadowed the capability-safety debate.
2020	**Language Models are Few-Shot Learners (GPT-3)** Tom Brown et al. (OpenAI)	175B parameters; in-context learning - tasks solved via prompts/examples with NO weight updates.	In-context learning is the property that turned LLMs into a general-purpose interface; launched the entire prompt-based ecosystem.
2020	**Scaling Laws for Neural Language Models** Jared Kaplan et al.	See domain 01 - power-law loss across N, D, C.	GPT-3's training budget was set by this curve; scaling became engineering.
2020	**Exploring the Limits of Transfer Learning? (T5: Text-to-Text Transfer Transformer)** Colin Raffel et al.	Cast every NLP task as text-to-text; systematic ablations of scale/data/Architectures.	The field's cleanest ablation study; unified interface precursor to instruction models.

Year	Paper / Authors	Core idea	Why it matters
2020	**Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (RAG)** Patrick Lewis et al.	Differentiable retrieval into a non-parametric memory (vector index) fused with generation.	Separates knowledge from weights - cheaper updates, fewer hallucinations, provenance; the backbone of enterprise LLM deployment.
2020	**The Curious Case of Neural Text Degeneration (nucleus sampling)** Ari Holtzman et al.	Greedy/beam text is degenerate; sampling from the nucleus (top-p mass) is human-like.	Sampling strategy is cognition-relevant: model distribution's tail matters; still the default decoding knob in every deployed LLM.
2021	**LoRA: Low-Rank Adaptation of Large Language Models** Edward Hu et al.	Freeze weights; learn low-rank update matrices - adapter quality at a fraction of memory.	Fine-tuning democratized (consumer GPUs); weight-deltas as shareable artifacts; the substrate of open-model customization.
2021	**RoFormer: Rotary Position Embedding (RoPE)** Jianlin Su et al.	Rotate query/key vectors by position-dependent angles - relative positions via complex-rotation structure.	The default positional scheme of modern LLMs; RoPE-scaling tricks (NTK, YaRN) drive long-context extension.
2022	**Training Compute-Optimal Large Language Models (Chinchilla)** Jordan Hoffmann et al.	See domain 01 - ~20 tokens/parameter at fixed compute.	Explained GPT-3-class models as undertrained; rewrote every lab's training recipe.
2022	**Training language models to follow instructions with human feedback (InstructGPT)** Long Ouyang et al. (OpenAI)	SFT on demonstrations -> RLHF (PPO) with a learned reward model from pairwise human preferences.	The RLHF pipeline; ChatGPT's recipe. Aligned models differ from base models through this stage, not pretraining.
2022	**Constitutional AI: Harmlessness from AI Feedback** Yuntao Bai et al. (Anthropic)	Replace human preference labels with AI feedback guided by an explicit written constitution + chain-of-thought critique.	RLAIF - supervision that scales with compute rather than human labelers; alignment-as-specified-principles.
2022	**Chain-of-Thought Prompting Elicits Reasoning in Large Language Models** Jason Wei et al. (Google)	Few-shot exemplars WITH intermediate steps dramatically unlock math/logic at large scale.	Reasoning as a prompting-recoverable ability; token-space computation begins here -> o1/R1.

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2022	**Large Language Models are Zero-Shot Reasoners (“Let’s think step by step”)** Takeshi Kojima et al.	A single zero-shot phrase triggers stepwise reasoning - no exemplars.	Reasoning capacity was already in the weights; prompting merely allocates tokens to it.
2022	**ReAct: Synergizing Reasoning and Acting in Language Models** Shunyu Yao et al.	Interleave chain-of-thought with tool/environment actions; reasoning plans actions, observations update reasoning.	The agent loop - nearly every deployed agent framework (tool use, browsing, code agents) is ReAct-shaped.
2022	**FlashAttention: Fast and Memory-Efficient Exact Attention** Tri Dao et al.	IO-aware exact attention - tiling that never materializes the N×N matrix.	Attention memory wall broken via systems thinking; enabled long-context training; template for kernel-level AI engineering (FlashAttention-2/3).
2022	**Switch Transformers: Scaling to Trillion Parameter Models** William Fedus, Barret Zohar, Noam Shazeer	Simplified top-1 MoE routing; trillion parameters with stable training.	MoE made routine; with Mixtral (2024) and DeepSeek-V3 (2024), sparse routing became the open-frontier default - capability per training-FLOP.

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2022	**In-Context Learning and Induction Heads** Catherine Olsson et al. (Anthropic)	Identifies induction heads - two-attention-head circuits implementing "copy pattern A->B, complete A->? with B" - as the mechanism behind much in-context learning; forms during a training phase change.	First circuit-level account of an emergent capability; mechanistic interpretability's flagship (with Nanda's grokking circuit, domain 01).
2022	**Toy Models of Superposition** Nelson Elhage et al. (Anthropic)	Networks store MORE features than neurons via nearly-orthogonal directions - polysemantic neurons are a consequence, not a bug.	Explains why interpretability is hard AND why scaling works (superposition <-> scaling laws, domain 01 F8/2025 mechanism); foundation of sparse-autoencoder feature research.
2023	**LIMA: Less Is More for Alignment** Zhou et al.	1,000 curated SFT examples on a strong base ≈ RLHF-level chat quality.	Alignment is mostly SURFACE behavior elicitation - the knowledge comes from pretraining; contradicting "alignment teaches capabilities."
2023	**Self-Consistency Improves Chain of Thought Reasoning** Xuezhi Wang et al.	Sample many reasoning paths, majority-vote answers.	Simplest test-time-compute scaling; the "ensemble over thoughts" ancestor of search-based reasoning.

Year	Paper / Authors	Core idea	Why it matters
2023	**Tree of Thoughts: Deliberate Problem Solving with LLMs** Shunyu Yao et al. (Princeton/DeepMind)	Explore a TREE of thought branches with LLM-guided search (BFS/DFS), value judgments, backtracking.	Explicit search over reasoning - MuZero's template in token space; big gains on planning-type tasks.
2023	**LLaMA: Open and Efficient Foundation Language Models** Hugo Touvron et al. (Meta)	Chinchilla-optimal small models (7–65B) trained on curated public data match much larger models.	Open weights + efficient recipe -> the entire open-ecosystem (Alpaca, Vicuna, Mistral lineage); weights-as-artifacts era.
2023	**Direct Preference Optimization (DPO)** Rafael Rafailov et al.	Optimize preferences directly with a classification-style loss - no reward model, no PPO.	RLHF-quality alignment with supervised-training simplicity; default in open post-training stacks.
2023	**Towards Monosemanticity: Decomposing Language Models With Dictionary Learning** Trenton Bricken et al. (Anthropic)	Sparse autoencoders extract monosemantic features (e.g., "Golden Gate Bridge", "code errors") from polysemantic layers.	Made features concrete and steerable (feature activation = behavior control); scaling monosemanticity (Templeton et al. 2024) extracted millions of features from Claude - interpretability became an engineering discipline.
2023	**Faith and Fate: Limits of Transformers on Compositionality** Nouha Dziri et al.	On multi-step compositional tasks transformers form parallel "shortcut" circuits whose error compounds exponentially with depth - they don't do serial abstraction natively.	Explains why chain-of-thought (externalizing steps into tokens) is necessary, not stylistic; motivates search and tool use.
2023	**GPT-4 Technical Report** OpenAI	Multimodal, scaled-up transformer; passes professional exams; (details withheld).	System-scale evidence of continued scaling returns + multimodality; began the closed-frontier era (benchmarks-first disclosure).

Year	Paper / Authors	Core idea	Why it matters
2024	**DeepSeek-V3 Technical Report** DeepSeek-AI	671B-total/37B-active MoE; Multi-head Latent Attention; FP8 training; 14.8T tokens for ~2.8M H800-GPU-hours (~$5.6M).	Frontier capability at 1/20th the cost of Western equivalents; engineering recipe (MLA, FP8, complementary-triangle MoE load balancing) now the open standard.
2024	**DeepSeekMath: Pushing the Limits of Mathematical Reasoning (GRPO)** Zhihong Shao et al.	Group Relative Policy Optimization - PPO without a separate critic; advantages from group-relative rewards.	The RL algorithm that made reasoning-RL cheap enough to run at scale; foundation of R1.

Year	Paper / Authors	Core idea	Why it matters
2024	**Scaling LLM Test-Time Compute Optimally Can Be More Effective than Scaling Model Parameters** Charlie Snell, Jaehoon Lee, Sanjeev Raghavan, Avirup Sil (UC Berkeley)	On hard math, a small model + optimized test-time search beats a 14× larger model.	Compute-optimal inference - a NEW scaling law axis; the theoretical spine of 2025's reasoning-model economics.
2025	**DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning** Daya Guo et al. (Nature, October 2025)	Pure RL with verifiable rewards (no SFT data, GRPO) elicits emergent reasoning behaviors - self-verification, reflection, long chains; test-time compute allocated by problem difficulty.	The defining result of 2025: reasoning is TRAINABLE BY RL ALONE at scale; R1-distill series moved open models to frontier reasoning; second scaling axis (inference compute) mainstreamed.
2025	**s1: Simple Test-Time Scaling** Muennighoff et al. (Stanford)	1K curated examples + "budget forcing" (extend/suppress "Wait" tokens) makes a 32B model rival far larger reasoning models.	Test-time compute control reduced to token-budget steering - embarrassingly effective; reasoning supervision needs are tiny.
2025	**The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models** Parshin Shojaee et al. (Apple)	On puzzle environments with controlled complexity, reasoning models' accuracy collapses beyond a complexity threshold; effort declines as problems near the frontier.	The strongest 2025 critique of "thinking" - pattern-matching vs. true search; frames the evaluation frontier (procedural, contamination-free benchmarks like ARC-AGI).
2025	**On the Biology of a Large Language Model (circuit tracing)** Anthropic Interpretability Team	Attribution graphs trace how individual prompts flow through features and circuits in Claude - planning in poems, multi-step reasoning observed directly.	(As of 2026) the state of the art in opening the black box; moves interpretability from "what's stored" to "how it computes" - the microscope for F9's emergence questions.
2025	**Advances and Open Problems for LLMs (survey)** (authors: see paper)	Analyzes 27 top models (2023–2025): reasoning saturation, data constraints, efficiency frontiers, evaluation crisis.	Best single map of the field's state as of early 2025 - use as the "what's actually open" reference.

Full sources with links live in the corpus: papers.md / papers.json per domain.

Where to go next

The living version of this canon lives in the corpus folder for this domain: papers.md carries every entry with full links and keywords, findings.md holds the complete distilled findings, and the diagrams folder contains Mermaid sources that render in Obsidian and GitHub. To see how this domain connects to the rest of the atlas, read the cross-domain map in 00_master-atlas; to extend the corpus, follow the conventions in AGENTS.md and re-run the index and PDF build scripts in tools/.